Indrajeet Singh

sihhhindra@gmail.com +91 8303550965
GitHub LinkedIn Coding Profile

Summary

Computer Science Engineering student at SRM IST with a 9.16 CGPA, focused on Data Structures & Algorithms, backend engineering and applied AI systems. Experienced in building relational database systems, RAG pipelines and LLM based applications using C++, Python, SQL and Docker. Currently strengthening Cloud infrastructure.

Education

SRM Institute of Science and Technology, Kattankulathur *Aug 2023 - May 2027*
Bachelor of Technology in Computer Science and Engineering **CGPA: 9.16**

Skills

- **Languages:** C++, Python, SQL
- **Database:** MySQL
- **AI / Data Science:** LangChain, LangGraph, RAG, Ollama, MCP, A2A, NumPy, Pandas
- **Cloud / DevOps / CI/CD:** AWS (EC2, S3, EBS, CloudWatch), Docker, Jenkins, Kubernetes
- **Tools:** Git, GitHub

Projects

Ensemble Learning Approach for Fraud Detection in Digital Payments - Python, KNN, Random Forest, Logistic Regression, SMOTE, PCA Link

- Built a machine learning system to identify potentially fraudulent digital payment transactions using multiple classification models.
- Combined KNN, Logistic Regression, and Random Forest models to improve fraud detection performance compared to using a single model.
- Handled class imbalance using SMOTE and applied PCA to reduce feature dimensions while retaining important information.
- Compared model performance using standard classification metrics and analyzed the strengths and limitations of individual models and the ensemble approach.

Low-Cost Multilingual Sentiment Analysis for Indic Languages - Python, NLP, Transformer Models, LoRA, Knowledge Distillation Link

- Developed a multilingual sentiment analysis system for Indic languages using transformer based language models.
- Used parameter efficient fine tuning with LoRA to adapt large language models while reducing training and computational requirements.
- Implemented knowledge distillation using a teacher-student approach to transfer knowledge from a larger multilingual model to a lightweight model.
- Evaluated the lightweight model for sentiment classification while focusing on reducing inference cost and maintaining useful performance.

MultiRAG - Content Aware Multimodal RAG System(Team Project with Madhav Tripathi - Python, LangChain, LangGraph, Ollama, Qwen, Docker Link

- Co-developed a local multimodal RAG system to answer questions from complex PDFs (text, tables, images, and diagrams), coordinating the ingestion and retrieval architecture via Git feature branches and pull requests.
- Designed a content aware ingestion pipeline utilizing semantic chunking, metadata tagging, table extraction, and vision based image understanding to parse document layouts.
- Implemented multimodal retrieval using cross-encoder reranking and local LLM/VLM inference to improve context relevance for generated answers.
- Evaluated the pipeline against a text only RAG baseline using an 'LLM as a judge' framework, achieving a 75% relative reduction in hallucination rate, and iterated on the architecture based on the findings.

Soft Skills

- **Collaboration & Teamwork:** Great at working with different teams, listening to everyone, and helping the whole group work toward the same goal.
- **Problem-Solving & Adaptability:** Ready for unexpected changes or problems, and quick to learn new tools and shift priorities when needed.
- **Team Alignment & SDLC:** Good at keeping developers and business stakeholders on the same page from the start of a project to the final launch.